

Levels of substitutional copper disturbed by an oxygen atom in n-type silicon

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ABSTRACT

The deep-level spectrum of n-type silicon crystals doped with copper during float-zone growth was investigated using the DLTS and Laplace-DLTS techniques. In addition to the well-known second acceptor of substitutional copper (Cu_s^-), two more bulk levels were observed after preliminary annealing at 350 °C: Cu_5 and Cu_6 at 0.16 and 0.59 eV below the bottom of the conduction band, respectively. The DLTS signatures of the Cu_5 and Cu_6 levels were practically identical to those of the $E_{\text{Cu}-5}$ and $E_{\text{Cu}-6}$ levels found by Markevich *et al.* in electron-irradiated, Cu-contaminated silicon [Solid State Phenom. **131-133**, 363 (2008)]. Considering our and literature data together, we infer that Cu_5 and Cu_6 are the second and first acceptor levels, respectively, of one defect. Accounting for the similarity of the Cu_5 and Cu_s^- levels, we presented experimental evidence showing that the Cu_5/Cu_6 defect is substitutional copper disturbed by a nearby oxygen atom, $\{\text{Cu}_s, \text{O}_i\}$. Hydrogen could be another constituent of the Cu_5/Cu_6 complex because the studied crystals were grown in wet argon atmosphere.

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I. INTRODUCTION

Transition metal impurities are dangerous in Si device fabrication technology.¹ In particular, copper (Cu), one of the fastest diffusers, has a relatively high solubility and forms ruinous precipitates in silicon crystals.²⁻⁴ The detection and study of copper-related defects are an imperative necessity to prevent copper precipitation.

An important issue is the interaction of copper with radiation defects, which are usually formed as a side effect of certain technological operations, such as ion implantation or plasma etching. Several groups have worked on modeling this situation. In a few studies, copper was introduced by deposition of Cu Schottky diodes on the electron-irradiated n- and p-type samples.^{5,6} An anomalously rapid disappearance of the deep levels associated with the divacancy (V_2) was observed in such diodes at temperatures below 150 °C. The authors attributed this to V_2 passivation due to a reaction with the mobile interstitial Cu_i atoms.

In another series of studies, copper was diffused into n-type Si by diffusion at 600–650 °C.^{7,8} Subsequent irradiation with 4 MeV electrons showed no effect of copper on the deep level spectrum,

and only standard radiation defects were observed. The presence of copper was only revealed during subsequent heat treatments at 150–250 °C through the enhanced annealing of the vacancy–oxygen pair (VO) and V_2 defects. This behavior was explained by an interaction of the radiation defects with mobile copper atoms released from unknown copper-containing agglomerates. However, in contrast to Ref. 5, the extinction of the radiation defects was accompanied by the emergence of two novel levels, $E_{\text{Cu}-5}$ and $E_{\text{Cu}-6}$ with activation energies of 0.17 and 0.60 eV, respectively. These levels were tentatively assigned to complexes of a copper atom with V_2 and VO, respectively.^{7,8}

However, these assignments raise some questions. Indeed, the assumption of an electrically active $\text{Cu}-V_2$ complex is in apparent contradiction with the results by Svensson *et al.*^{5,6} As for the copper–VO interaction, the donor and acceptor levels of the CuVO complex were later found in the lower half of the bandgap.^{9,10} These results are discussed in more detail in Sec. IV.

Below we show that two levels denoted as Cu_5 and Cu_6 can be found in n-type Si crystals doped with copper during growth. The DLTS signatures of Cu_5 and Cu_6 are very close to those of

$E_{\text{Cu}-5}$ and $E_{\text{Cu}-6}$. However, the crystals used in this study were never irradiated with energetic particles and therefore had no VO and V_2 centers. The identity and origin of the Cu_5/Cu_6 ($E_{\text{Cu}-5}/E_{\text{Cu}-6}$) levels are discussed in the light of the entire body of data on copper-related defects in Si crystals. It is concluded that the presence of hydrogen introduced during crystal growth may play an important role.

II. EXPERIMENTAL DETAILS

The silicon single crystals used in this work were grown by float-zone technique (FZ-Si) in an argon atmosphere.¹¹ Copper doping was achieved by dissolving a predetermined amount of the metal in the molten zone. The conductivity type was determined by doping polysilicon rods with either phosphorus or boron ($\sim 10^{15} \text{ cm}^{-3}$ in both cases). For our experiments, 1-in. wafers of $\sim 1 \text{ mm}$ thick were cut from the 15–20 cm long boules.

Some wafers were annealed in the temperature range of 350–650 °C for 30 min in air. The anneals were terminated with a fast (few seconds) cooling by dropping the specimens on a metal block. Prior to further investigations, all as-received and annealed wafers were subjected to chemical etching in a mixture of HF and HNO_3 acids, which removed $\sim 30 \mu\text{m}$ from the surface.

Schottky diodes were prepared by gold or aluminum vacuum deposition through a shadow mask. The ohmic contacts were made with InGa paste. The deep-level spectra of the crystals under study were investigated using the standard DLTS, Laplace-DLTS (L-DLTS),¹² and CV techniques, with a 1 MHz test signal being used in all cases. Most of the DLTS curves were taken with a filling pulse duration of 1 ms and a rate window of 49 s^{-1} . Further details on the sample characterization can be found elsewhere.¹³

III. RESULTS

A. Preliminary anneals

The crystal growth parameters were targeted to obtain a concentration of about 10^{14} cm^{-3} of substitutional copper atoms (Cu_s).¹⁴ However, the DLTS measurements on as-received wafers did not reveal any peaks related to the Cu_s levels, which should appear at 107 K in n-type samples and at 103 and 220 K in p-type samples for the rate window used (Fig. 1).^{13,15,16} The relatively weak peaks of unknown origin observed in n-type at 90 and 205 K correspond to a defect concentration of $\sim 3 \times 10^{12} \text{ cm}^{-3}$. The M peak in p-type samples corresponds to a Cu-related center often observed in the copper-diffused Si crystals.¹⁷ The absence of Cu_s levels is attributed to the fact that all Cu_s atoms are incorporated into the Cu_{PL} complexes of four copper atoms^{18,19} with a single level near the valence band.^{20–22} Indeed, the spectrum of the as-received p-type samples was dominated by the Cu_{PL} peak with an amplitude approximately corresponding to the expected Cu_s concentration [Fig. 1(b)]. The presence of Cu_{PL} centers in the as-received n-type samples was confirmed by the observation of the characteristic photoluminescence band with the zero-phonon line at 1014 meV.²³

The Cu_{PL} center exhibits a moderate thermal stability and can be destroyed by annealing at 200–250 °C.^{16,23} Only Cu_s , the backbone of Cu_{PL} , remained after these treatments. It should be

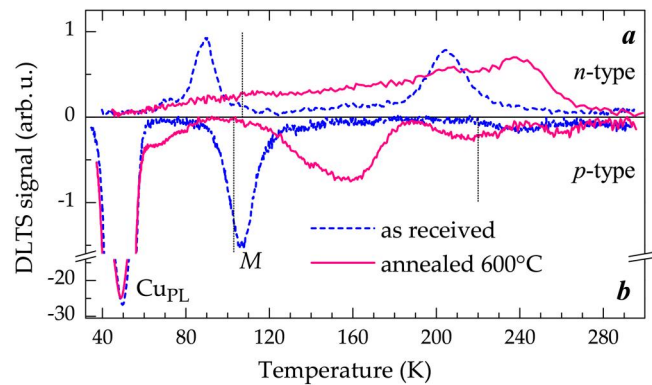


FIG. 1. The DLTS spectra of as-received (dashed curves) and annealed at 600 °C (solid curves) Cu-doped Si crystals of n-type (a) and p-type (b). The spectra were taken from layers at ~ 1.7 – $3.7 \mu\text{m}$ depths with a 49 s^{-1} rate window. The vertical dotted lines indicate the expected position of the Cu_s peaks.

noted that these results were obtained from samples in which copper was introduced by diffusion. However, in our crystals, which were doped with copper during growth and had a significantly higher overall copper concentration,¹⁴ this treatment left behind some Cu_{PL} complexes. Increasing the annealing temperature to 600–650 °C caused all of the Cu_s centers to convert back to Cu_{PL} (Fig. 1). Such heat treatments most likely lead to the dissociation of unknown copper-containing agglomerates and the formation of a high concentration of mobile interstitial copper atoms (Cu_i). Therefore, the Cu_{PL} complexes form again after annealing due to the decoration of the Cu_s centers with the Cu_i species.

It was found experimentally that annealing at 350–400 °C destroys the Cu_{PL} centers and simultaneously removes the mobile Cu_i species. A typical DLTS spectrum in the n-type wafer annealed at 350 °C for 30 min is shown by the solid curve in Fig. 2. In general, the spectrum looks similar to those previously reported for copper-doped n-type Si and is dominated by the peak at $\sim 105 \text{ K}$, which could be related to the copper second acceptor level.^{15,16} The defects that determine the E_{145} and E_{185} peaks are located in the near-surface layer and were formed during chemical etching before the Schottky diode formation.²⁴ (The subscript in the E_{xxx} notation indicates the temperature of the DLTS peak at a rate window of 49 s^{-1} .)

B. Room-temperature peak

The main visual difference from previously reported spectra on similar samples is the Cu_6 peak at $\sim 305 \text{ K}$. Despite the unusually high temperature, the peak exhibited properties typical of a normal deep level. The peak amplitude remained practically constant both when the filling pulse duration was varied in the range of 0.3–3 ms and when the setup rate window was set to 7.9 s^{-1} , resulting in a shift of the peak maximum down to 284 K.

The electron emission rate from the Cu_6 level was measured using the Laplace-DLTS (L-DLTS)¹² technique in the temperature

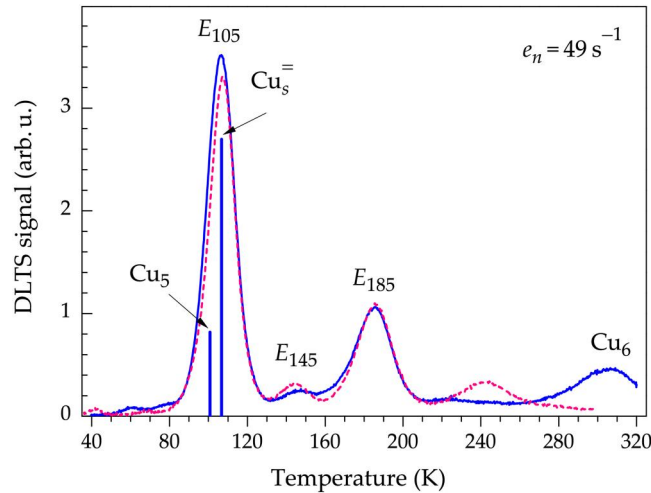


FIG. 2. The DLTS spectra of an n-type Cu-doped Si crystal measured after annealing at 350 °C (solid curve) and after successive treatments at 600 and 350 °C (dashed curve). The signals were taken from the layers at 1.6–2.8 μm depth.

range of 284–320 K, and the results are plotted with diamonds in Fig. 3(a). The activation energy was found to be 585 ± 4 meV, while the emission rate at 300 K averaged over all studied samples was 31.0 s^{-1} . The dashed line in Fig. 3(a) was copied from Ref. 7 ($E_{\text{Cu}-6}$ level). This close correlation suggests that the same level is

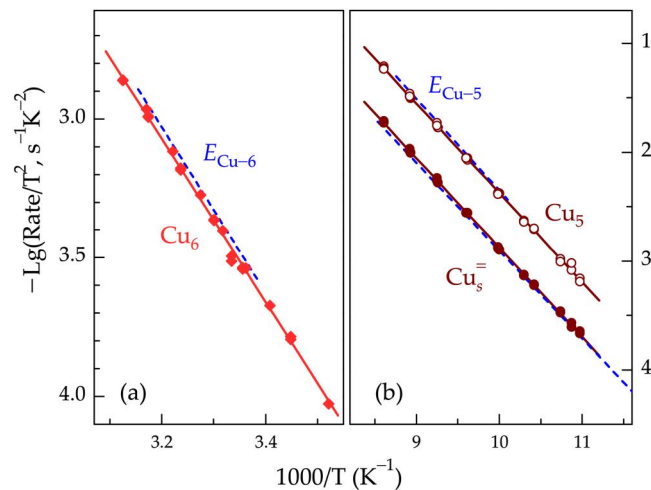


FIG. 3. Arrhenius plots for the levels observed in n-type Cu-doped Si after annealing at 350 °C at room temperature (a) and around 100 K (b) (see Fig. 2). Symbols represent the emission rates measured in two separately prepared samples for each level, and the solid lines are the best fits to the experimental points. Dashed lines are the literature data from Ref. 7 ($E_{\text{Cu}-5}$ and $E_{\text{Cu}-6}$) and from Ref. 16 (Cu_s).

observed in our work. We retain both notation options below (Cu_6 and $E_{\text{Cu}-6}$) to make it clear whether we are discussing our data or the results obtained on irradiated samples.^{7,8}

Since the Cu_6 level is close to the midgap, one would also expect a DLTS signal in p-type samples, as was observed for the gold acceptor level in silicon, for example, Ref. 25. Although no systematic studies on this issue have been performed, the only DLTS curve taken with a low rate window of 7.9 s^{-1} in a similarly grown p-type crystal demonstrated a peak at 284 K, i.e., exactly where the Cu_6 signal appears in n-type crystals. The amplitude of this peak was ~ 5 times smaller than the largest Cu_6 signal ever observed in n-type samples.

C. Doublet at 105 K

The L-DLTS analysis at lower temperatures revealed that the peak at ~ 105 K is determined by two levels, the electron emissions of which differ by three times approximately (solid curve in Fig. 4). The temperature positions and amplitudes of the corresponding peaks are schematically shown by the vertical lines in Fig. 2.

Figure 3(b) shows the Arrhenius plots measured for these two levels in the temperature range of 91–116 K (symbols). Activation energies for the Cu_s and Cu_5 levels were found to be 165 ± 2 and 163 ± 3 meV, respectively. The average emission rates at 100 K were equal to 12.8 and 41.7 s^{-1} , respectively. The dashed lines in Fig. 3(b) were copied from the literature. The line from Ref. 16 confirms that the slower component is indeed determined by the Cu_s level. Comparison of the points for faster emission with the dashed line taken from Ref. 7 ($E_{\text{Cu}-5}$) suggests that the same defect is observed in the present work and justifies the Cu_5 notation.

Deep levels can also be characterized by how carrier emission rates vary with electric field strength. Such dependencies for the Cu_5 and Cu_s levels are shown in Fig. 5. The L-DLTS signals were taken from a layer of $\sim 0.3 \mu\text{m}$ width located about $1.7 \mu\text{m}$ below the Schottky diode (for methodology, see Ref. 26). In the whole

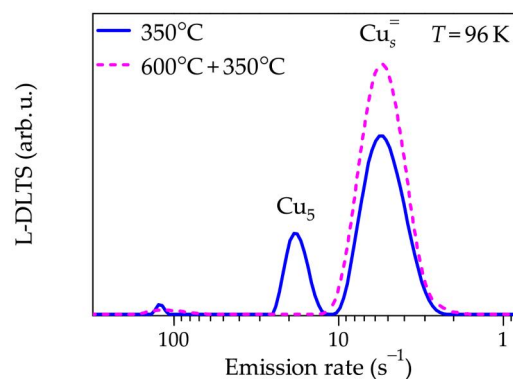


FIG. 4. Laplace-DLTS spectra of Cu-doped Si after 350 °C annealing of the as-received crystal (solid curve) and of the sample pre-annealed at 600 °C (dashed curve). (Note the inverted abscissa scale.) The spectra were taken at 96 K from layers at 2.4–3.3 μm depth under the average electric field of 1.7 V/ μm .

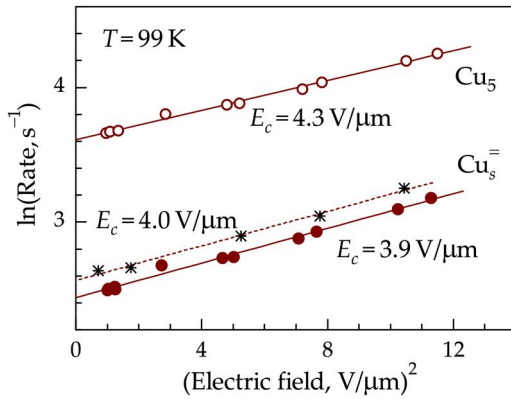


FIG. 5. The electron emission rates at 99 K from the Cu_5 and Cu_s^- levels as a function of electric field. The data are from the as-received crystal after 350 °C annealing (open and filled circles) and from the sample successfully annealed at 600 and 350 °C (stars).

range of electric field studied ($1.0 \leq \mathcal{E} \leq 3.4 \text{ V}/\mu\text{m}$), the emission rates from both levels depend exponentially on the square of the electric field $R(\mathcal{E})/R_0 = \exp(\mathcal{E}^2/\mathcal{E}_c^2)$. Such dependence is usually related to the phonon-assisted tunneling.^{27–29} The critical fields $\mathcal{E}_c$ measured for the Cu_5 and Cu_s^- levels (4.3 and 3.9 $\text{V}/\mu\text{m}$ at 99 K, respectively) are close to those predicted by theory.^{28,29}

D. Depth distribution

The depth profiles of main levels are shown in Fig. 6. The points for the Cu_5 and Cu_s^- levels (open and filled circles) were

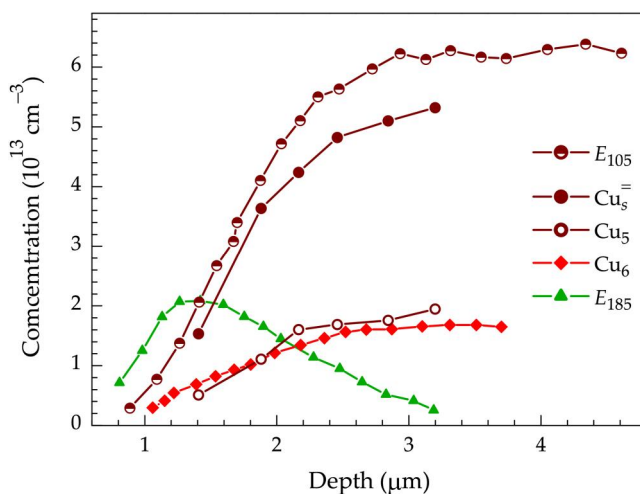


FIG. 6. Depth profiles of the deep levels in an n-type Cu-doped Si crystal after the 30 min annealing at 350 °C and subsequent chemical etching. The E_{105} , E_{185} , and Cu_6 data were obtained using the standard DLTS profiling. The Cu_5 and Cu_s^- profiles were measured by the L-DLTS technique.

calculated from the L-DLTS measurement, with the bias and pulse amplitudes adjusted to keep the average electric field in the tested layer below 1.7 $\text{V}/\mu\text{m}$ in all cases. These two depth profiles have almost identical shapes: The average point-to-point $[\text{Cu}_5]/[\text{Cu}_s^-]$ ratio is 0.36 with a standard deviation of $\sim 5\%$. For other samples prepared in the same way, this ratio was in the range of 0.35–0.49, but the profile shapes always remained very similar.

Due to the limitations of the L-DLTS setup, the depth profiles could only be extended to 3.2 μm in our samples. Therefore, the standard DLTS setup was used for profiling in a wider range. The half-filled circles in Fig. 6 show the profile measured at 105 K. Up to 3.2 μm , the E_{105} values are in good agreement with those expected for the sum signal from the Cu_5 and Cu_s^- levels. At greater depths, E_{105} reaches a plateau, suggesting that Cu_5 , like Cu_s^- , is a bulk defect.

The depth profile of the Cu_6 level is shown with diamonds in Fig. 6. These points were taken at 284 K using the standard DLTS setup with the 7.9 s^{-1} rate window. The plateau at greater depths suggests that the Cu_6 level also belongs to a bulk defect.

The decrease in concentration of all detected centers in the near-surface region is most likely due to passivation by hydrogen injected into the sample during chemical etching prior to Schottky diode deposition. This view is supported by the fact that the concentrations of both passivated phosphorus atoms (calculated from the CV measurements) and deactivated E_{105} centers decay exponentially into the sample bulk with the same characteristic length of about 0.4 μm .

E. Thermal stability of the Cu_5 and Cu_6 levels

To check thermal stability of the observed defects, the as-received wafers (as well as the samples after the 350 °C treatment) were annealed at 500–650 °C with fast cooling. As shown in Fig. 1, such treatment resulted in a weak structure-less DLTS spectrum in n-type samples. In the p-type crystals, annealing at ~ 600 °C also resulted in a weak deep-level spectrum except for the strong Cu_{PL} signal. Therefore, an additional 350 °C annealing was required to reveal the Cu_s^- levels.

The dashed curve in Fig. 2 shows the deep-level spectrum of the n-type sample consequently annealed at 600 and 350 °C. In general, the spectrum is similar to that measured after the first 350 °C annealing of the as-received wafer. The most obvious distinction is the absence of the Cu_6 peak. (After annealing at 500 °C, an order of magnitude weaker Cu_6 signal was observed in some cases.) The L-DLTS studies also showed no Cu_5 signal (Fig. 4). It is important for further discussion that the concentration of Cu_s^- levels after 600 °C annealing is nearly equal to the sum concentration of Cu_s^- and Cu_5 before annealing (see Figs. 2 and 4).

IV. DISCUSSION

The DLTS signatures of the Cu_5 and Cu_6 levels are very close to those of the $E_{\text{Cu}-5}$ and $E_{\text{Cu}-6}$ levels found in irradiated copper-contaminated Si^{7,8} (Fig. 3). It is highly unlikely that the similarity of the two pairs of the Arrhenius DLTS signatures is coincidental, especially given that all levels were found only in copper-doped crystals. Therefore, we infer that the Cu_5 and Cu_6 levels in our crystals are the same as $E_{\text{Cu}-5}$ and $E_{\text{Cu}-6}$, respectively, and the following discussion is based on this identity.

A. Cu_5 vs Cu_s^-

The electron emission rates from the Cu_5 and Cu_s^- levels are governed by identical (within experimental accuracy) activation energies and similarly depend on the strength of electric field (see Sec. III C). We also tried to distinguish between the Cu_5 and Cu_s^- levels by applying shorter filling pulses during the DLTS measurements. With a pulse duration of $0.2 \mu\text{s}$, the amplitude of the E_{105} peak was reduced by a factor of 15 relative to its maximum value, which corresponds to a capture cross section of $\sim 4 \times 10^{-17} \text{ cm}^2$. However, the shape of the E_{105} DLTS peak remained unchanged, indicating that the electron capture rates for the Cu_5 and Cu_s^- levels are very similar.

Note also that the depth profiles of the Cu_5 and Cu_s^- levels have always almost identical shapes (Fig. 6). This means that these defects react in the same way with hydrogen atoms that penetrate into the near-surface layer during chemical etching.

All these observations suggest that the defect that defines the Cu_5 level is closely related to Cu_s . This point of view is also supported by the appearance of additional Cu_s^- levels after the annealing of Cu_5 at 600°C (Figs. 2 and 4). More specifically, one could assume that Cu_5 is the second acceptor level of a Cu_s atom that is slightly perturbed by an unknown defect.

Understanding the identity of the Cu_5 and $E_{\text{Cu}-5}$ levels forces us to reevaluate the results obtained from the copper-saturated and electron-irradiated Si samples.^{7,8} Markevich *et al.* linked the $E_{\text{Cu}-5}$ level to a $\text{Cu}-\text{V}_2$ complex. However, this complex does not resemble the desired Cu_s -like defect. A closer look at Fig. 4 in Ref. 8 reveals that the $E_{\text{Cu}-5}$ concentration continues to increase due to annealing at $200\text{--}250 \text{ K}$, even after the V_2 signal disappears. In addition, the electrical activity of the $\text{Cu}-\text{V}_2$ complex contradicts other studies that reported the divacancy energy levels were completely passivated due to interaction with copper.^{5,6,9,30}

An alternative possibility is that the $E_{\text{Cu}-5}$ level is associated with the center resulting from a $\text{Cu}_i\text{--VO}$ reaction. This assumption is fully compatible with the experimentally observed kinetics of transformation of the deep-level spectra during isochronal annealing.^{7,8} The theory predicts that the defect resulting from the $\text{Cu}_i\text{--VO}$ interaction is actually a substitutional copper atom with oxygen bridging two adjacent Si atoms, $\{\text{Cu}_s, \text{O}_i\}$.³¹ Indeed, the CuVO center was observed in p-type, electron-irradiated Si crystals contaminated with copper.⁹ The two levels of this center differ only barely from the Cu_s donor and acceptor levels. Therefore, based on the results from electron-irradiated samples, it is suggestive to identify $E_{\text{Cu}-5}$ with the second acceptor level of the CuVO defect.

B. Cu-O complex formation in FZ-Si

The correlation of $E_{\text{Cu}-5}$ (Cu_5) with the $\{\text{Cu}_s, \text{O}_i\}$ defect raises questions about the formation of such pairs in our non-irradiated crystals doped with copper during growth. The Cu_5 level was observed after the first annealing at 350°C , but it was destroyed when the temperature increased above 500°C (Fig. 4). This means that the Cu_5 centers were formed in our crystals during the post-growth cooling below 500°C or during long-term storage at room temperature. A simple estimate shows that forming the $\{\text{Cu}_s, \text{O}_i\}$ pair requires one of the components to diffuse $\sim 10^{-5} \text{ cm}$.

To our knowledge, there are no data on the diffusivity of substitutional copper in the literature. The calculations show that the reaction $\text{Cu}_i + \text{V} \rightarrow \text{Cu}_s$ releases 3.12 eV .²² Apparently, the reverse reaction is practically forbidden at temperatures below 500°C . The only conceivable mechanism for Cu_s diffusion is the vacancy mechanism, which requires free vacancies. The formation energy of an additional vacancy in a perfect Si crystal is $\sim 4 \text{ eV}$.^{22,32} Recent *ab initio* calculations showed that the vacancy formation energy is reduced in the vicinity of copper-rich complexes.²² A deeper analysis reveals that the main energy gain comes from the subsequent $\text{Cu}_i + \text{V} \rightarrow \text{Cu}_s$ recombination. However, no increase of the Cu_s concentration was observed in our experiments.

An alternative way to form the $\{\text{Cu}_s, \text{O}_i\}$ pairs is through oxygen mobility. Obviously, the diffusivity of isolated O_i atoms is too low by many orders of magnitude, and no significant amount of oxygen dimers is expected in our oxygen-lean FZ-Si. However, it is known that oxygen diffusivity below 500°C can be enhanced by hydrogen.^{33,34} Our samples have a relatively high hydrogen concentration because they were grown in a wet argon atmosphere.¹¹

The phenomenon of hydrogen-accelerated oxygen diffusion was discovered long ago,^{33,34} yet it remains poorly understood.³⁵ It is unclear what the oxygen mobility will be at a given temperature and hydrogen concentration $[\text{H}]$. An experimental attempt³⁶ to determine the lower limit of $[\text{H}]$ required for accelerated thermal donor formation yielded values that are orders of magnitude lower than the estimated hydrogen concentration $[\text{H}] \geq 10^{14} \text{ cm}^{-3}$ in our samples.¹¹

The Cu_5 and Cu_6 levels do not recover with repeated annealing at 350°C after being destroyed at 600°C (Figs. 2 and 4). Given the mechanism of $\{\text{Cu}_s, \text{O}_i\}$ pair formation discussed above, the most natural explanation for this result is that annealing at 600°C also causes hydrogen to diffuse out of the sample. The diffusion length of isolated hydrogen atoms indeed exceeds the thickness of the wafers used for a given thermal budget at 600°C .^{35,37}

C. Cu_5/Cu_6 center

Markevich *et al.* found two Cu-related levels in irradiated Si.^{7,8} We found *both* these levels in quite differently Cu-doped crystals. It is highly unlikely that Cu_5 and Cu_6 are not related. Moreover, we present arguments below in favor of these two levels belonging to the same center.

In all of our experiments, the Cu_5 and Cu_6 levels appeared and disappeared simultaneously. The concentrations of both levels were similar in all the samples studied, although the concentration of Cu_6 was systematically somewhat lower, by no more than 20% (Fig. 6). This aligns with the observation of a weak Cu_6 signal in p-type crystals (Sec. III B).

The $E_{\text{Cu}-5}$ and $E_{\text{Cu}-6}$ levels were originally assigned to different centers.^{7,8} However, thorough analysis reveals strong correlations between these two levels. Figure 2(b) in Ref. 8 shows how DLTS spectra develop in irradiated, copper-contaminated Si with isochronal annealing. Unfortunately, the spectra were cut at 300 K , preventing an immediate estimate of the $E_{\text{Cu}-6}$ concentration. From the spectra extended to a higher temperature (Fig. 3 in Ref. 8), one can see that the $E_{\text{Cu}-6}$ peak maximum is ~ 1.9 times higher than the DLTS signal at 300 K . Using this factor, the $[E_{\text{Cu}-5}]/[E_{\text{Cu}-6}]$

concentration ratio was found to be nearly constant at 1.2 over the wide temperature range, from the appearance of these signals after annealing at 175 °C until heating to 275 °C. After annealing at higher temperatures, the assumption that the $E_{\text{Cu}-5}$ and $E_{\text{Cu}-6}$ levels belong to one center seems to be disproved, as the amplitudes of the low- and high-temperature peaks differ significantly (Fig. 3 in Ref. 8). However, deconvolution of the measured DLTS spectra into ideal peaks revealed that the higher signal at ~ 100 K is due to the appearance of an additional peak.³⁸ This peak resembles the signal from the second acceptor level of Cu_5 .

Assigning the Cu_5 and Cu_6 levels to the same center raises questions again about the nature of this defect. The Cu_5 level may be the second acceptor level of the CuVO center, as it is not documented in the literature. However, the Cu_6 level cannot be associated with the CuVO acceptor level, as the latter was observed in p-type crystals.⁹ Although the $\text{CuVO}^{-/0}$ from Ref. 9 and Cu_6 levels are close within the bandgap, the corresponding DLTS peaks appear at different temperatures (224 and 305 K, respectively, for the rate window of 49 s^{-1}). This rules out the possibility that these peaks are manifestations of the same level. Therefore, the CuVO model for the Cu_5/Cu_6 center is not valid.

However, the CuVO center is not the only possible configuration of $\{\text{Cu}_5, \text{O}_i\}$ pair. Annealing at ≥ 300 °C resulted in the transformation of CuVO into a Cu_5^* center with levels that are virtually identical to those of Cu_5 .⁹ This transformation was attributed to the partial dissociation of the CuVO center. Estimates showed that the thermal budget was insufficient for the full release of the oxygen atom, but a few diffusion jumps weakened its effect on the Cu_5 levels. However, Cu_5^* cannot be correlated with Cu_6 for the same reasons as CuVO .

A recent study³⁹ provided another example of how an oxygen atom can occupy different positions in the vicinity of a Cu-related defect. At least three configurations of the $\{\text{Cu}_{\text{PL}}, \text{O}_i\}$ complex were found to interconvert at temperatures between 180 and 260 °C. Therefore, one can assume that the Cu_5 and Cu_6 levels belong to another configuration of the $\{\text{Cu}_5, \text{O}_i\}$ pair, different from CuVO and Cu_5^* . Since CuVO forms at room temperature and Cu_5^* requires annealing above 300 °C,⁹ the Cu_5/Cu_6 configuration is likely an intermediate stage of gradual $\{\text{Cu}_5, \text{O}_i\}$ dissociation.

Another possible assumption is that the Cu_5/Cu_6 center includes a hydrogen atom. As discussed above, the presence of hydrogen is a prerequisite for the formation of $\{\text{Cu}_5, \text{O}_i\}$ pairs in our oxygen-lean crystals. Although all models of hydrogen-accelerated oxygen diffusion assume low O–H binding energy,^{40–42} hydrogen can remain in the complex through direct bonding to copper, $\{\text{Cu}_5\text{H}, \text{O}_i\}$. The stability of the $\text{Cu}_5\text{–H}$ bond up to 500 °C has been both theoretically predicted^{31,43} and experimentally confirmed.¹³ This is consistent with the annealing of the Cu_5/Cu_6 center at 500–600 °C. Regarding the irradiated samples,^{7,8} the presence of hydrogen is confirmed by the strong peak at ~ 170 K (Fig. 2 in Ref. 8), which is determined by the VOH acceptor level.^{44,45}

V. CONCLUSION

We studied the deep-level spectrum of the n-type Si crystals doped with copper during growth. In addition to the dominant second acceptor level of substitutional copper (Cu_s^-), two more

levels were observed in the bulk of the wafers after annealing at 350 °C. These are the Cu_5 and Cu_6 levels located 0.16 and 0.59 eV below the bottom of the conduction band, respectively. The DLTS signatures of these levels are the same as those of two previously discovered levels resulting from the interaction between interstitial copper and radiation defects: $E_{\text{Cu}-5}$ and $E_{\text{Cu}-6}$, respectively.^{7,8} Finding the same levels in crystals with such different defect histories greatly narrows the range of possible defect models.

Our data and the revision of the results obtained from irradiated samples⁸ lead us to conclude that the Cu_5 and Cu_6 levels belong to the same center. Furthermore, the results listed in Sec. IV A indicate that this is a Cu_5 -like defect. The only way to create such a defect in irradiated samples is through the $\text{Cu}_i + \text{VO}$ reaction. Therefore, we inferred that the defect is Cu_5 perturbed by a nearby oxygen atom, $\{\text{Cu}_5, \text{O}_i\}$. Two configurations of such pair have been observed previously in irradiated p-type crystals.⁹ However, the Cu_6 level cannot be associated with either of these configurations. Therefore, it can be assumed that the Cu_5/Cu_6 defect corresponds to an additional $\{\text{Cu}_5, \text{O}_i\}$ configuration.

There is also the subtle question of how the $\{\text{Cu}_5, \text{O}_i\}$ pair forms in our oxygen-lean crystals. Since crystals were grown in a wet argon atmosphere and contain large amounts of hydrogen,¹¹ we hypothesize that hydrogen-enhanced oxygen diffusion occurs at temperatures below 500 °C during ingot cooling after growth. Additionally, the presence of hydrogen allows us to propose an altered defect model and assign Cu_6 and Cu_5 to the first and second acceptor levels of the $\{\text{Cu}_5\text{H}, \text{O}_i\}$ complex.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Nikolai Yarykin: Conceptualization (equal); Data curation (lead); Investigation (lead); Writing – original draft (lead); Writing – review & editing (equal). **Jörg Weber:** Conceptualization (equal); Funding acquisition (lead); Project administration (equal); Resources (lead); Supervision (lead); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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